

Methods: GLEAM_J041509-282854

ALMA Thesis Planner (automated pipeline)

0.1 0. Before the plan is fixed: archive reconnaissance and provenance

Your first job is not analysis, it is finding out exactly what exists. Query the ALMA Science Archive for project code 2023.1.01342.S, both through the web portal and programmatically with ‘astroquery.alma’, and build a delivery table: which targets have delivered executions, how many execution blocks each, the proprietary status and whether your PI-team credentials give you access, the array configuration, the delivered spectral-window centres and bandwidths, and the shortest and longest baselines actually present in each execution block. Do not populate the frequency column from the receiver-band definitions. ALMA Band 1 and Band 3 have nominal receiver coverages of roughly 35–50 GHz and 84–116 GHz, but those are observed-frame receiver-band definitions, not this project’s tuning; the frequencies you carry through the whole thesis must be read off the delivered SPW headers with ‘listobs’ (and from the ‘casa_pipescript.py’ and ‘flux.csv’ files in the QA2 weblog). Record the aggregate continuum effective frequency per band per target, because that is what your spectral indices and surface brightnesses are defined at.

In the same pass, extract the per-execution absolute flux-scale uncertainty as quoted in each QA2 report — do not substitute a remembered rule of thumb — and note the CASA version the pipeline ran under, because you will pin that version for any restore.

Second Step Zero item: distances. Physical-scale conversion is the whole thesis, so the distance per galaxy must come from the source the PI’s target list itself used, and you record it in a ‘targets.csv’ with three columns — value, uncertainty, and provenance string. If the target list does not record distances, ask the PI for that provenance; do not substitute a catalogue of your own choosing. For nearby galaxies, redshift-derived distances carry peculiar-velocity error at the tens-of-percent level, and that error propagates directly onto the compactness axis, so it must be a tracked quantity and not a footnote.

If Step Zero shows the data are not accessible to you, bring that to me immediately. It is a conversation about the project, not something to engineer around.

0.2 1. Sample definition

Fix the sample before you image anything. Take the turnover/control classification as given from the proposal: it is your independent variable and you never adjust it. From the sixteen targets, select approximately five turnover galaxies and approximately five controls matched pairwise in distance and in angular size, using the distances from ‘targets.csv’ and the angular sizes from the target list. Write the matching rule down explicitly (for example, pair each turnover galaxy with the control minimising the joint fractional offset in distance and major-axis angular diameter) and freeze the pairing in a file. Sixteen targets across two bands is the time sink that would otherwise sit ahead of every science result; additional targets are added only after the full chain — imaging

→ common beam → missing-flux audit → size and surface brightness → comparison plot — has been driven end to end once on the matched ten.

0.3 2. Retrieval, restoration, and continuum imaging

Download the pipeline-calibrated measurement sets and the QA2 continuum images and weblogs. Work from the delivered products by default: restore the calibrated data with ‘scriptForPI.py’ under the pinned CASA version, and re-image yourself only where a delivered product is inadequate for your purpose (wrong field of view, no matched-taper version available, an obvious artefact). Self-calibration is off the critical path — if a bright target tempts you, do it as a clearly labelled side branch and never let it feed the headline numbers.

Before continuum imaging, identify and flag line-contaminated channels at each target’s redshift. Redshift the relevant rest-frame line frequencies into the observed frame using each target’s own redshift and check them against your delivered SPW coverage: in the Band 3 range these include CO(1–0) at 115.271 GHz rest, CS(2–1) at 97.981 GHz rest, HCN(1–0) at 88.632 GHz rest and HCO⁺(1–0) at 89.189 GHz rest; in the Band 1 range, CS(1–0) at 48.991 GHz rest, hydrogen radio recombination lines, and the class I methanol and SiO maser transitions near 36, 43 and 44 GHz rest. Confirm empirically with ‘plotms’ amplitude-versus-frequency on the calibrated, continuum-dominated data, then flag generously — you are protecting a continuum measurement, not measuring lines.

Image with ‘tclean’, ‘specmode=’mfs’, ‘deconvolver=’mtmfs’ with ‘nterms=2’ given the fractional bandwidth, and a fixed Briggs robust value used identically for every target and band. Keep a per-image log of the achieved beam, the residual RMS, the cleaning threshold and the mask used.

0.4 3. The two common beams

This is the methodological backbone, and I want you to be very clear in your own head that there are *two distinct* homogenisations, done in this order.

Common angular beam, within a galaxy, across bands. The two bands differ by roughly a factor of 2.5 in frequency and hence in native resolution, and the delivered configurations differ target to target. Homogenise in the *uv* plane first, not just in the image plane: compute the *uv* distributions in units of $k\lambda$ (not metres) for both bands of a given target, then re-image both with a common ‘uvrange’ selection — inner limit set by the larger of the two minimum baselines, outer limit by the smaller of the two maxima — plus a ‘uvtaper’ chosen so the two dirty beams match. Then force the final match in the image plane by restoring both with an identical ‘restoringbeam’, or by post-convolving with ‘imsmooth(targetres=True)’. Verify the match by differencing the restored beams, not by eye.

Common physical beam, across galaxies. Now convert. For each galaxy the physical beam is $\theta \times D$ with D from ‘targets.csv’. Find the worst physical resolution in the sample and degrade every other galaxy to it, which means a *different* angular beam per galaxy. Only after this step are two galaxies’ sizes and surface brightnesses comparable at all. Do the convolution with ‘imsmooth’ / ‘radio-beam’, and record the achieved physical beam in pc, with its uncertainty inherited from the distance uncertainty, in the results table. If you skip or fudge this, a resolution difference will masquerade as the compactness signal you are hunting, and the thesis is worthless.

0.5 4. Maximum recoverable scale and the missing-flux audit

Compute the maximum recoverable scale from the *actual* shortest baselines in each delivered execution, not from the proposal’s configuration request. Extract the baseline distribution with ‘msmd’ or analysisUtils, state explicitly which convention you adopt ($\theta_{\text{MRS}} \approx 0.6 \lambda/L_{\text{min}}$, or the ALMA Technical Handbook’s fifth-percentile-baseline definition — pick one, justify it, and use it everywhere), and tabulate θ_{MRS} per band per target alongside that target’s angular size.

Then quantify recovery rather than merely flagging risk. Inject mock sources into a copy of each measurement set with CASA’s ‘ft’ — circular Gaussians of known total flux spanning a grid of intrinsic FWHM from well below the beam to a few times θ_{MRS} — and image them through the *identical* tclean call, common-beam convolution included. Measuring recovered-over-injected flux versus intrinsic size gives you a per-target, per-band flux-recovery curve. That curve is what converts “we might be resolving out flux” into a bound: for each galaxy, given its angular size, you can either correct the measured flux or quote an explicit bracket on the resolved-out fraction. This matters doubly because resolving out diffuse emission biases the sample toward apparent compactness — precisely the direction of the hypothesis — so the audit’s purpose is to demonstrate that any turnover-versus-control difference is not an artefact of differing uv coverage, angular size, or distance. Plot $\theta_{\text{MRS}}/\theta_{\text{source}}$ for turnover and control galaxies side by side and confirm the two subsamples are not systematically offset.

0.6 5. Photometry

Measure integrated flux densities in apertures matched between bands and defined on the common-physical-scale images: a fixed physical radius aperture (chosen once, from the sample’s coarsest physical beam) plus a curve-of-growth out to where the profile flattens. Use ‘photutils’ on the exported FITS, or CASA ‘imstat’ with region files; either is fine, but use one consistently. The flux uncertainty combines the residual-map RMS scaled by the number of independent beams in the aperture, added in quadrature with the per-execution absolute flux-scale uncertainty from that target’s QA2 report, and the two contributions are reported separately as well as combined, since the calibration term is correlated within an execution and the noise term is not.

0.7 6. The compactness measurement

Per band and per galaxy, at the common beam and in physical units, produce:

- **Size.** Fit a single elliptical Gaussian with CASA ‘imfit’ and take the beam-deconvolved FWHM, converted to pc. Adopt a quantitative resolvability criterion based on the deconvolved size relative to the beam and the source signal-to-noise (the standard Condon-style minimum resolvable size), and where the source fails it, report an upper limit on size — which is a *lower* limit on surface brightness, and therefore points in the informative direction for the hypothesis. Also report the non-parametric half-light radius R_{50} from the curve of growth, which does not assume a Gaussian.
- **Surface brightness.** Peak and aperture-mean, converted from Jy/beam to Rayleigh–Jeans brightness temperature and to MJy/sr (both beam-explicit), and then to a physical surface brightness in luminosity per pc^2 using the target distance.
- **Concentration.** The fraction of flux inside a fixed physical radius, $f(<R_{\text{fix}})$, with R_{fix} set once for the whole sample and large enough to exceed the coarsest common physical beam. Treat this as the primary compactness statistic: it is the most robust of the three to beam

and missing-flux effects, because both numerator and denominator are measured in the same map at the same resolution.

Propagate uncertainties by Monte Carlo: resample the flux calibration, the distance (using its recorded uncertainty), and the pixel noise, re-run the size/concentration extraction on each realisation, and take the resulting distributions as your error bars. Do not propagate distance error analytically — it enters the physical beam and the physical aperture simultaneously and the covariance matters.

0.8 7. Pre-stated detectability, and blinding

Before you look at the comparison plot with the labels attached, write down — from the achieved common physical beam, the distance spread, and the flux-recovery bounds — the **minimum size ratio and minimum surface-brightness ratio** the data could distinguish between the two subsamples. Derive these by simulation: draw two mock subsamples of five from your actual measured uncertainties, impose a known contrast, and find the contrast at which your test rejects the null in, say, 80% of trials. Commit that number to the thesis text and to the repository, timestamped, before unblinding.

Keep the analysis blind operationally: run the whole chain on anonymised target IDs, with the turnover/control key in a separate file that you merge only at the final plotting step.

With ~ 5 per subsample, use nonparametric tests — a two-sided Mann–Whitney U or an exact permutation test on the difference in medians — and quote the effect size with a bootstrap confidence interval rather than leaning on a p-value. Because several galaxies will yield size upper limits, handle the size comparison with survival-analysis methods for left-censored data (Kaplan–Meier estimation and a Gehan or logrank two-sample test, e.g. via ‘lifelines’); the concentration statistic, which is not censored, remains your cleanest test. The headline result is sizes, surface brightnesses and concentrations plotted for turnover versus control galaxies with missing-flux bounds attached, and interpreted against the pre-stated thresholds — so that an overlap reads as “compact structures of the kind required are absent above this contrast level.”

0.9 8. Thermal fraction, as an explicitly illustrative quantity

Compute the two-point spectral index between the delivered Band 1 and Band 3 effective frequencies from your matched-aperture, common-beam fluxes. Then state plainly that one spectral index cannot separate three components: free-free dominates at the low end, dust rises at the high end, and synchrotron is already weak at Band 1. No infrared data are available here to pin the dust, and GLEAM’s optically-thin end cannot pin the synchrotron slope for precisely the turnover galaxies of interest, because there it is contaminated by the absorption under study. So report thermal fractions as illustrative only, computed on a declared grid: a fiducial non-thermal slope, a declared dust contribution parameterised by a modified blackbody with stated β , free-free with $S \propto \nu^{-0.1}$, and $T_e = 10^4$ K (the free-free emissivity depends only weakly on T_e). Show the spread across the grid and state that the systematic is likely comparable to or larger than any turnover-versus-control difference. Nothing in the headline result may depend on this decomposition.

0.10 9. Secondary descriptive product

Within the turnover subsample only, plot surface brightness and concentration against the GLEAM turnover frequency taken from the MWA GLEAM photometry the abstract names, and quote a

Spearman rank coefficient. Present this as a description of the sample, not as a test.

0.11 10. Scope control and reproducibility

If imaging throws a surprise, the single cut is the per-region resolved analysis of individual compact structures in the best-resolved targets; galaxy-integrated sizes, concentrations and upper limits stay. The sample is already the small matched set, so no further cut is available.

Keep everything in a git repository: pinned CASA version, the tclean and imsmooth calls as scripts rather than notebook cells, ‘targets.csv’ with distance provenance, the flux-recovery curves as data products, and a plain-text data journal recording every decision and its date. Every table in the thesis should be regenerable by running one script. Include the required acknowledgement of ADS/JAO.ALMA#2023.1.01342.S and the standard NRAO acknowledgement in any resulting publication.

Schedule. ~6 weeks: retrieval, CASA training, reproduce one QA2 image end to end. ~8 weeks: imaging, common-beam convolution and photometry for the matched sub-sample. ~6 weeks: maximum-recoverable-scale and missing-flux audit plus the pre-stated detectability thresholds. ~6 weeks: sizes, surface brightnesses, concentrations and the subsample comparison, adding further targets here if time permits. ~6 weeks: resolved extension and writing.